

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: MLA0303

COURSE NAME: Reinforcement learning

COURSE OUTCOME COVERED

CO4: Evaluate model-based reinforcement learning approaches for prediction, planning, and policy optimization. (BL3, BL4, BL5)

Assessment Tool Weightage: 40%

Dr . DUMPALA PRASANTH

QUESTIONS: 25

Total Marks: 50

Annexure A

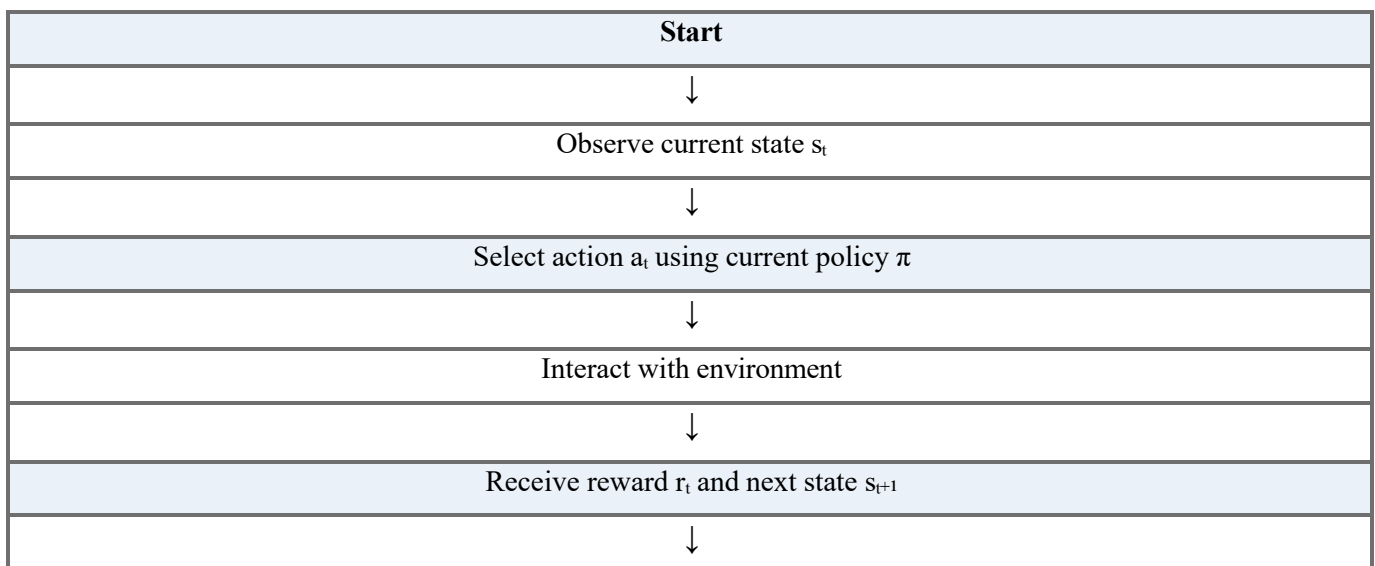
Design and Develop Model

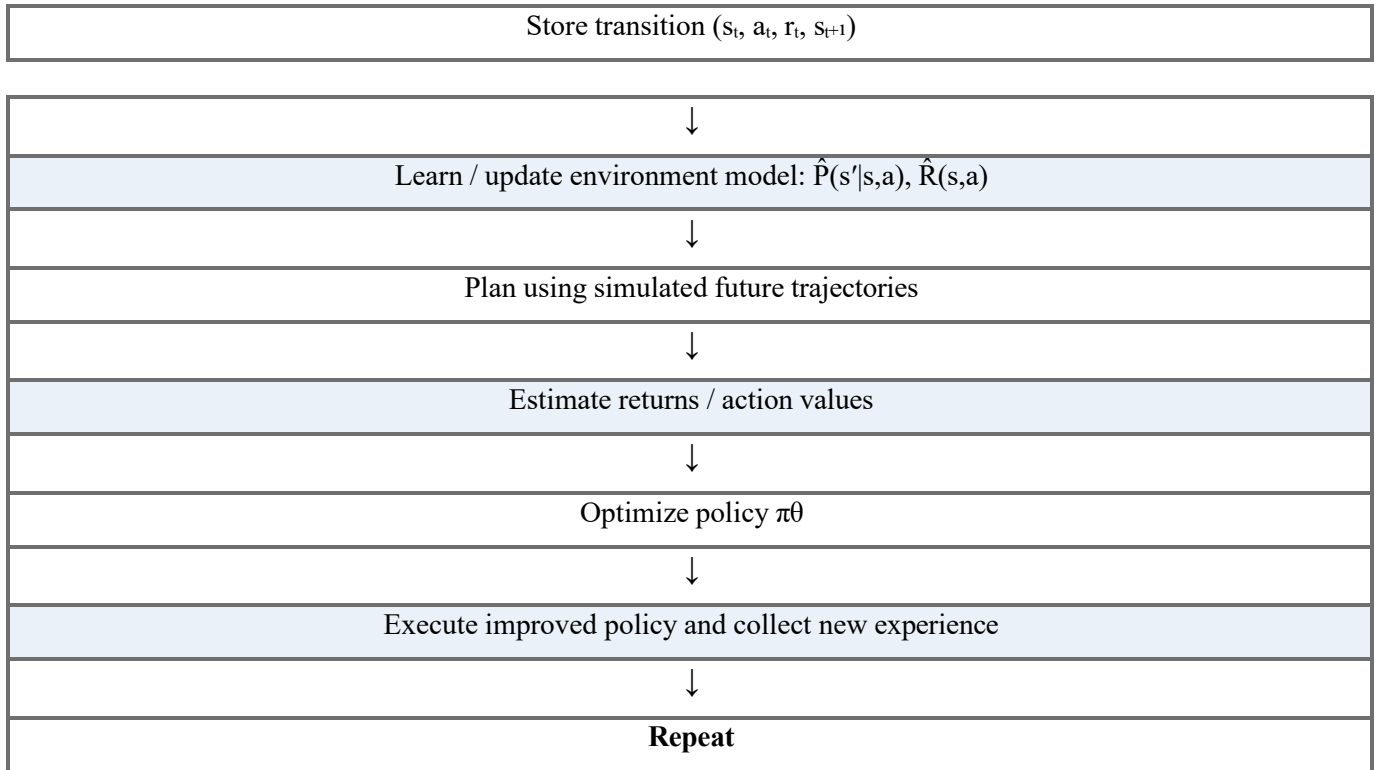
DURATION: 4 HRS (2 Sessions)

Model-Based RL Approach

1. Draw a flowchart illustrating the complete workflow of a Model-Based Reinforcement Learning agent from environment interaction to policy optimization.

Answer:





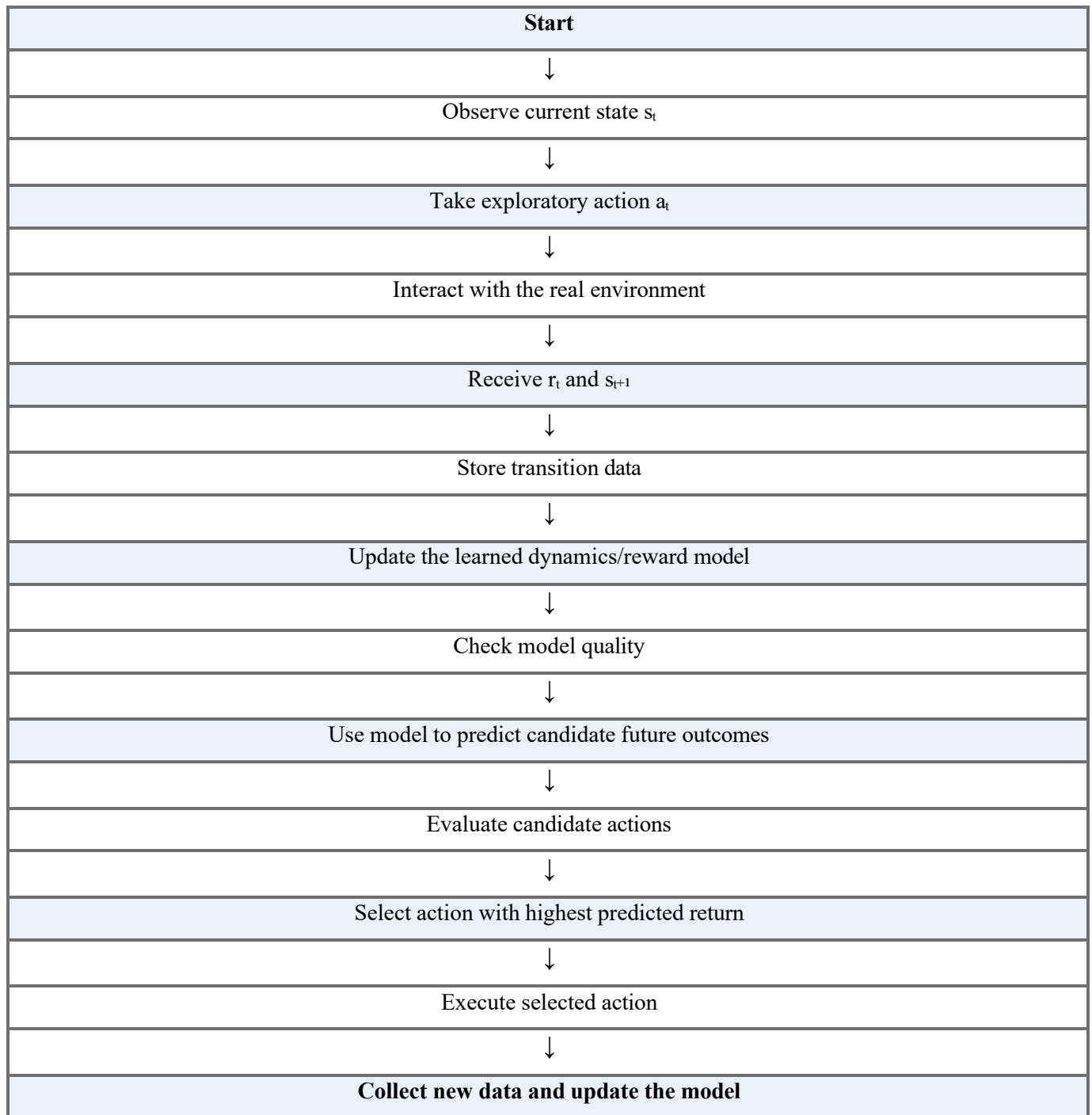
2. Draw a concept map showing the relationship between the Environment, Model, Planning Module, Policy, and Reward in Model-Based Reinforcement Learning.

Answer:

MODEL-BASED REINFORCEMENT LEARNING	
Environment	Provides states, transitions and rewards; receives selected actions.
Model	Learns/predicts transition dynamics and rewards from experience.
Planning Module	Uses the learned model to simulate futures and compare actions.
Reward	Measures the quality of actions and provides the learning objective.
Policy	Selects actions and is improved using planning/value information.
Main loop	Environment → experience → model → planning → policy → environment

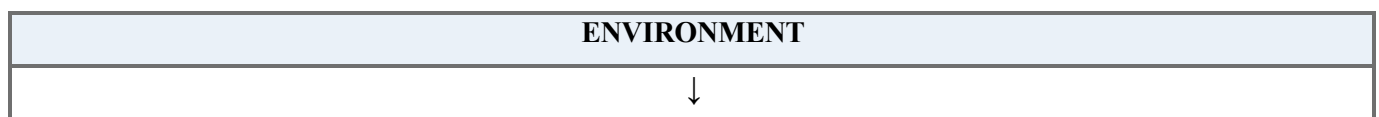
3. Design a flowchart explaining how an RL agent learns an environment model before selecting an optimal action.

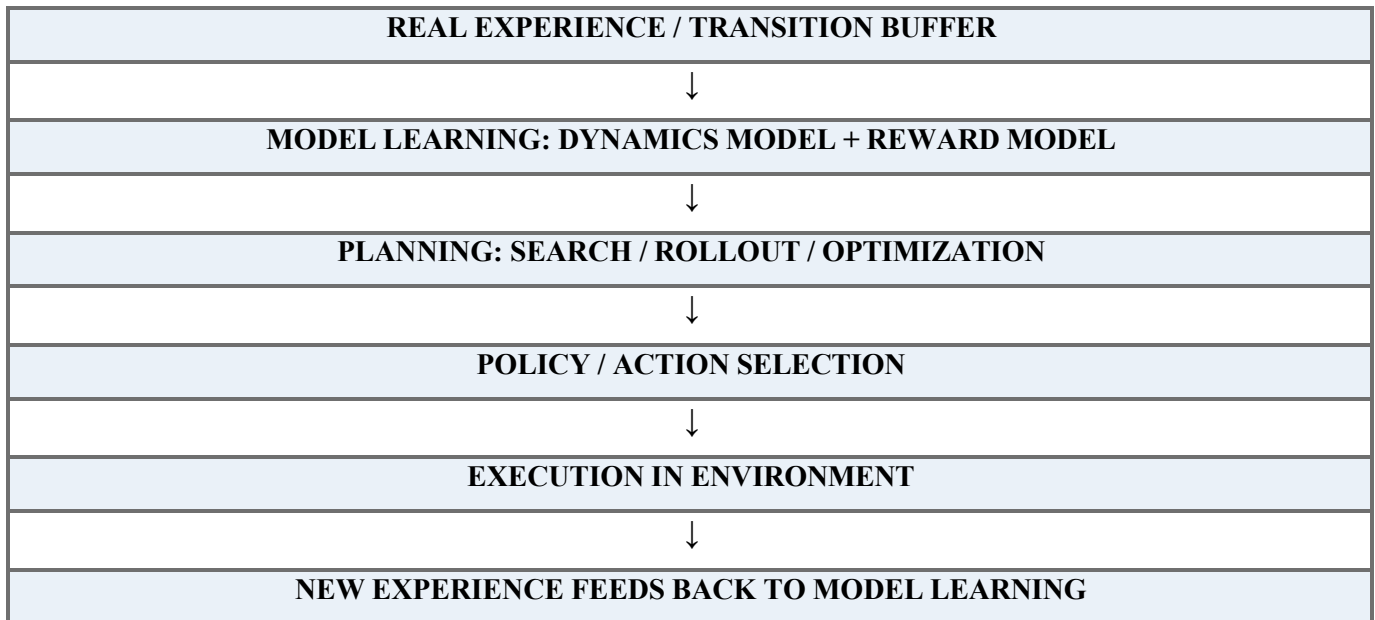
Answer:



4. Draw the architecture of a Model-Based RL system showing model learning, planning, and execution components.

Answer:





5. Prepare a concept map comparing Model-Based Reinforcement Learning and Model-Free Reinforcement Learning.

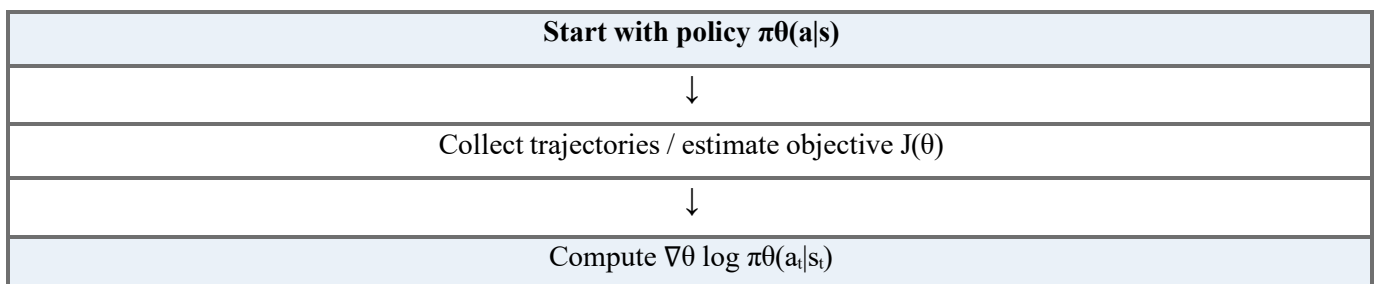
Answer:

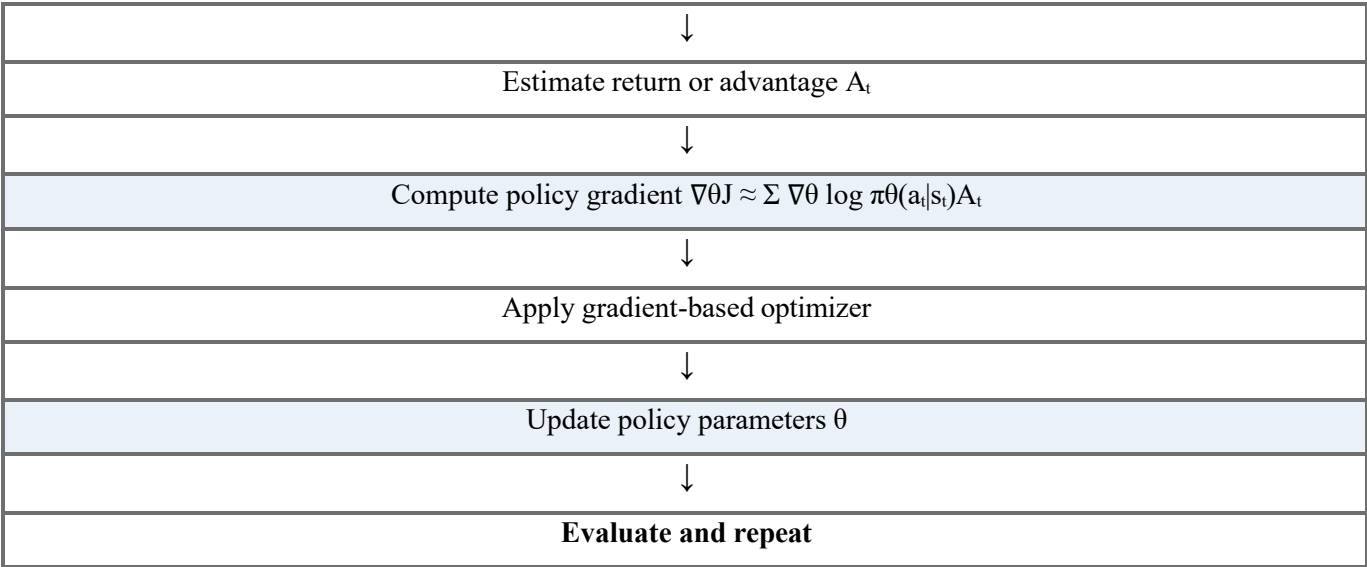
MODEL-BASED RL	MODEL-FREE RL
Learns/uses an explicit environment model.	Learns policy/value directly from experience.
Uses planning or simulated rollouts.	Normally does not require an explicit dynamics model.
Can be more sample-efficient.	Often needs more real interaction data.
Sensitive to model bias and prediction error.	Avoids explicit model-error problems.
More computation may be spent on planning.	Computation is concentrated on direct policy/value learning.

Analytic Gradient Computation

6. Draw a flowchart explaining the steps involved in analytic gradient computation for policy optimization.

Answer:





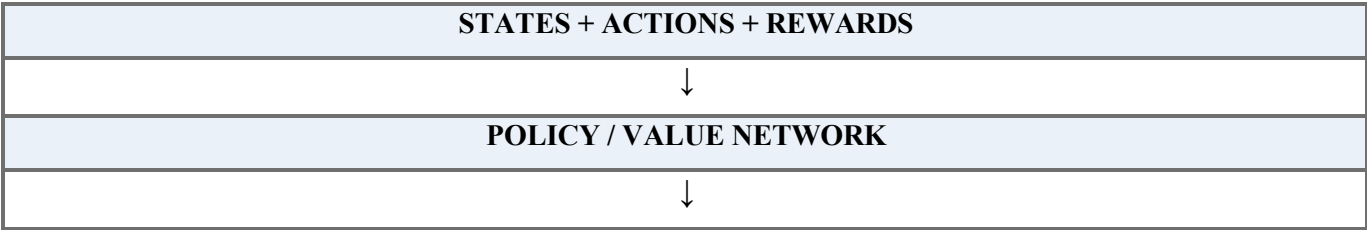
7. Design a concept map showing how gradients are computed and propagated during policy learning.

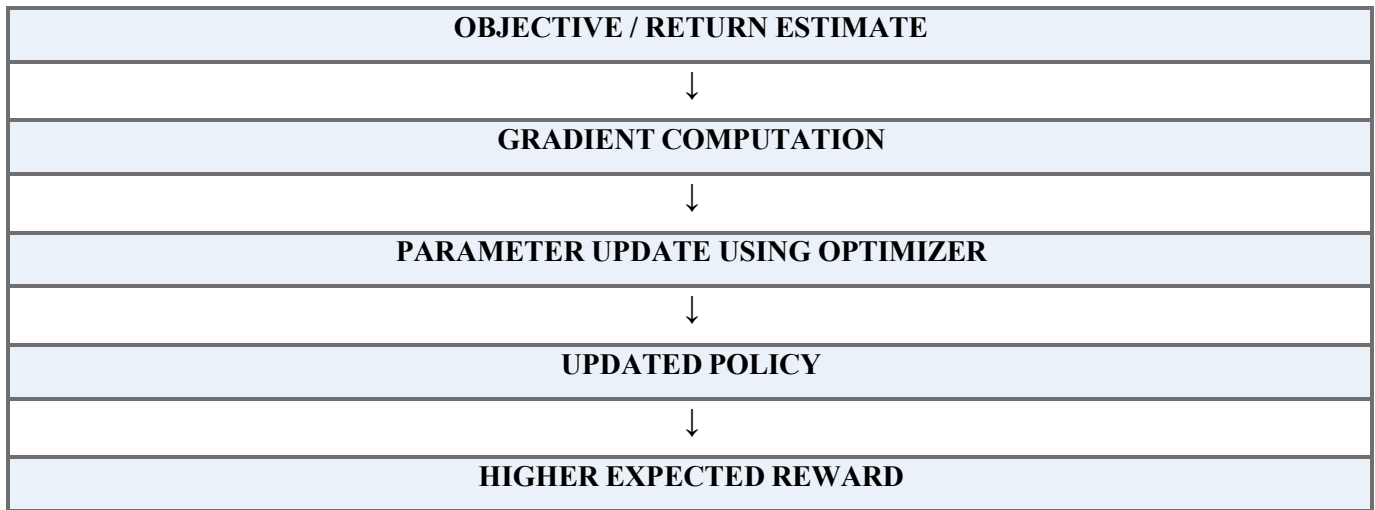
Answer:

POLICY LEARNING WITH GRADIENTS	
Policy Parameters θ	Determine the policy $\pi \theta(a s)$.
Trajectory Data	Provides states, actions and rewards.
Objective $J(\theta)$	Measures expected return.
Gradient $\nabla \theta J$	Shows how parameters should change to improve the objective.
Backpropagation	Propagates derivatives through the computation graph.
Optimizer	Uses gradients to update θ .
Updated Policy	Produces new behavior and new experience.

8. Draw a block diagram illustrating gradient computation, parameter updates, and reward optimization in Reinforcement Learning.

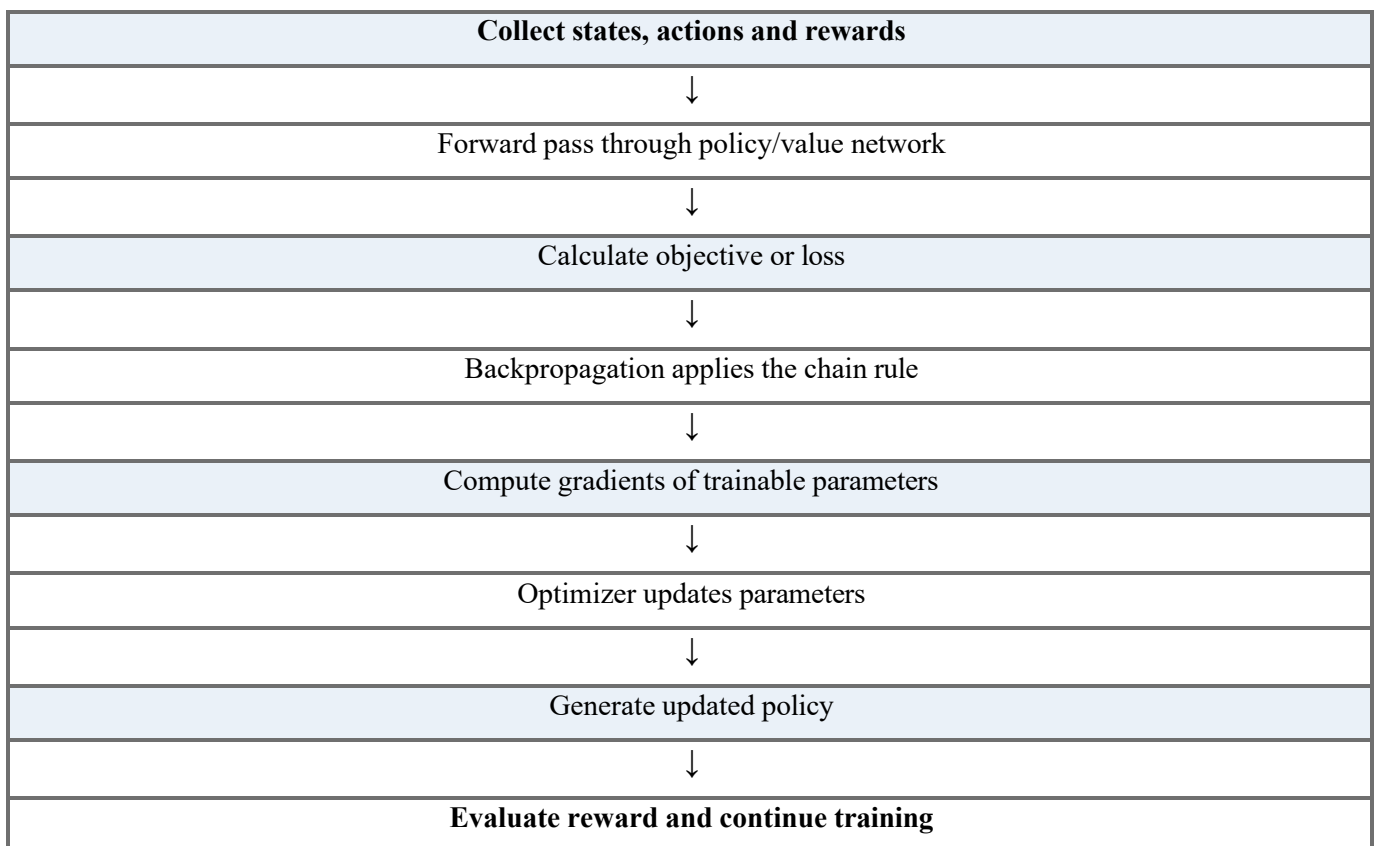
Answer:





9. Prepare a flowchart explaining gradient-based optimization using backpropagation in Reinforcement Learning.

Answer:



10. Draw a concept map showing the relationship among policy parameters, gradients, optimization algorithms, and rewards.

Answer:

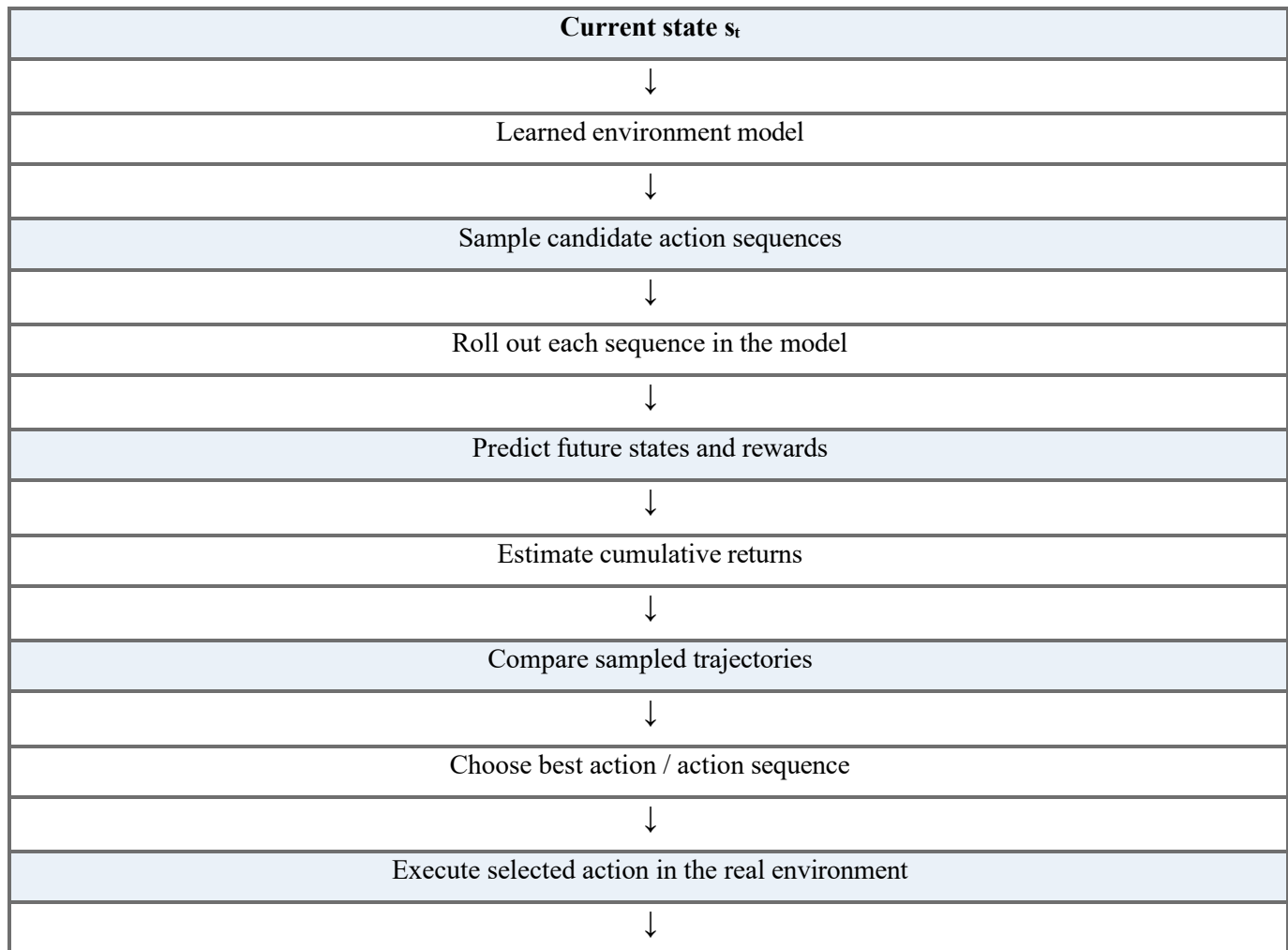
POLICY PARAMETERS → GRADIENTS → OPTIMIZATION → REWARDS

Policy Parameters θ	Control the behavior/action probabilities.
Rewards	Provide feedback on action quality.
Objective $J(\theta)$	Converts reward information into a learning target.
Gradients $\nabla \theta J$	Indicate the direction and magnitude of useful parameter change.
Optimization Algorithm	SGD/Adam applies the gradient with a learning rate.
Updated θ	Changes the policy and influences future rewards.

Sampling-Based Planning

11. Draw a flowchart illustrating the sampling-based planning process in Model-Based Reinforcement Learning.

Answer:



Use new transition to improve the model

12. Prepare a concept map showing how sampled trajectories are generated and evaluated for planning.

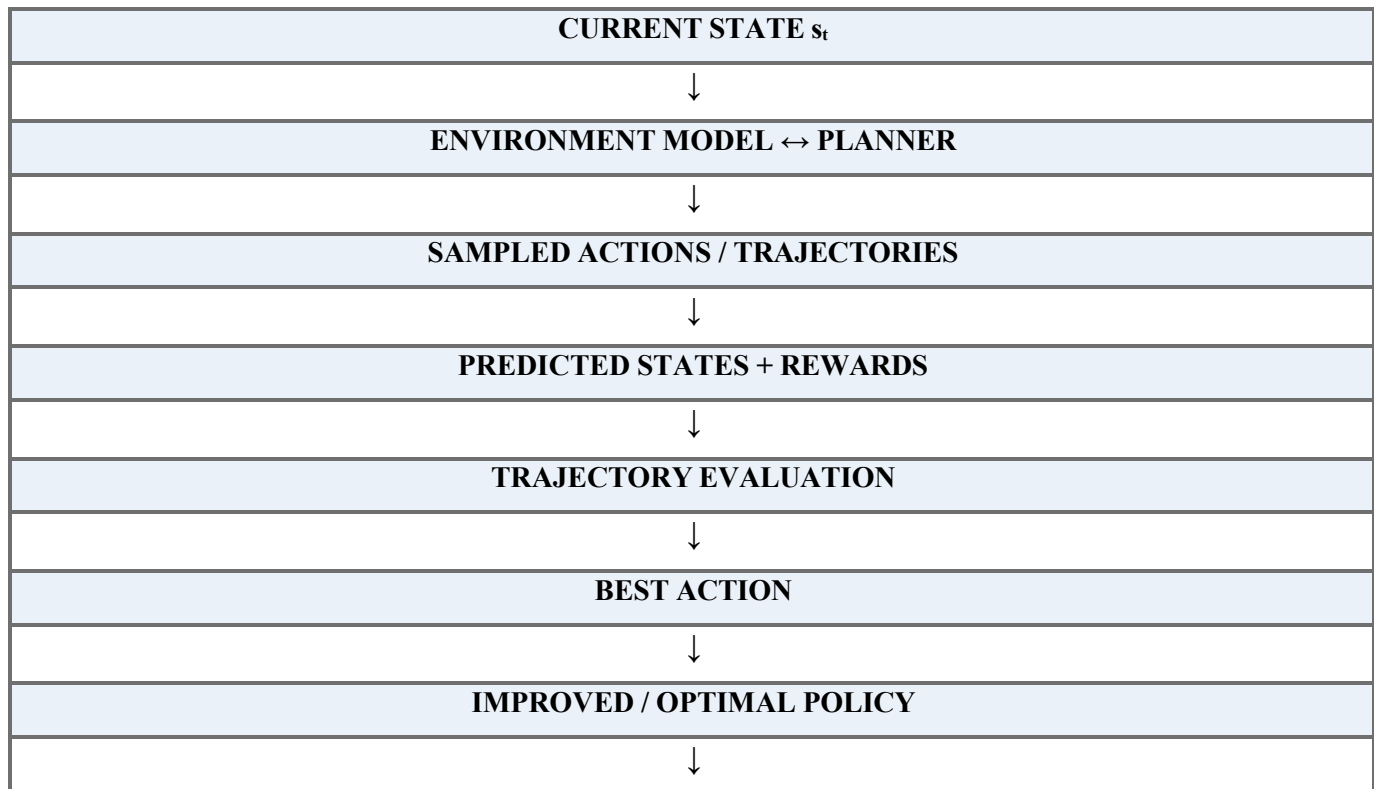
Answer:

SAMPLED TRAJECTORY EVALUATION

Current State	Starting point for simulated trajectories.
Sampled Actions	Generate alternative action sequences.
Environment Model	Predicts next states and rewards.
Sampled Trajectories	Contain predicted states, actions and rewards.
Return Evaluation	Discounted cumulative rewards score each trajectory.
Planner	Compares scores and selects the promising action.
Policy	Uses the planning result in the real environment.

13. Draw a block diagram showing the interaction between the environment model, planner, sampled actions, and optimal policy.

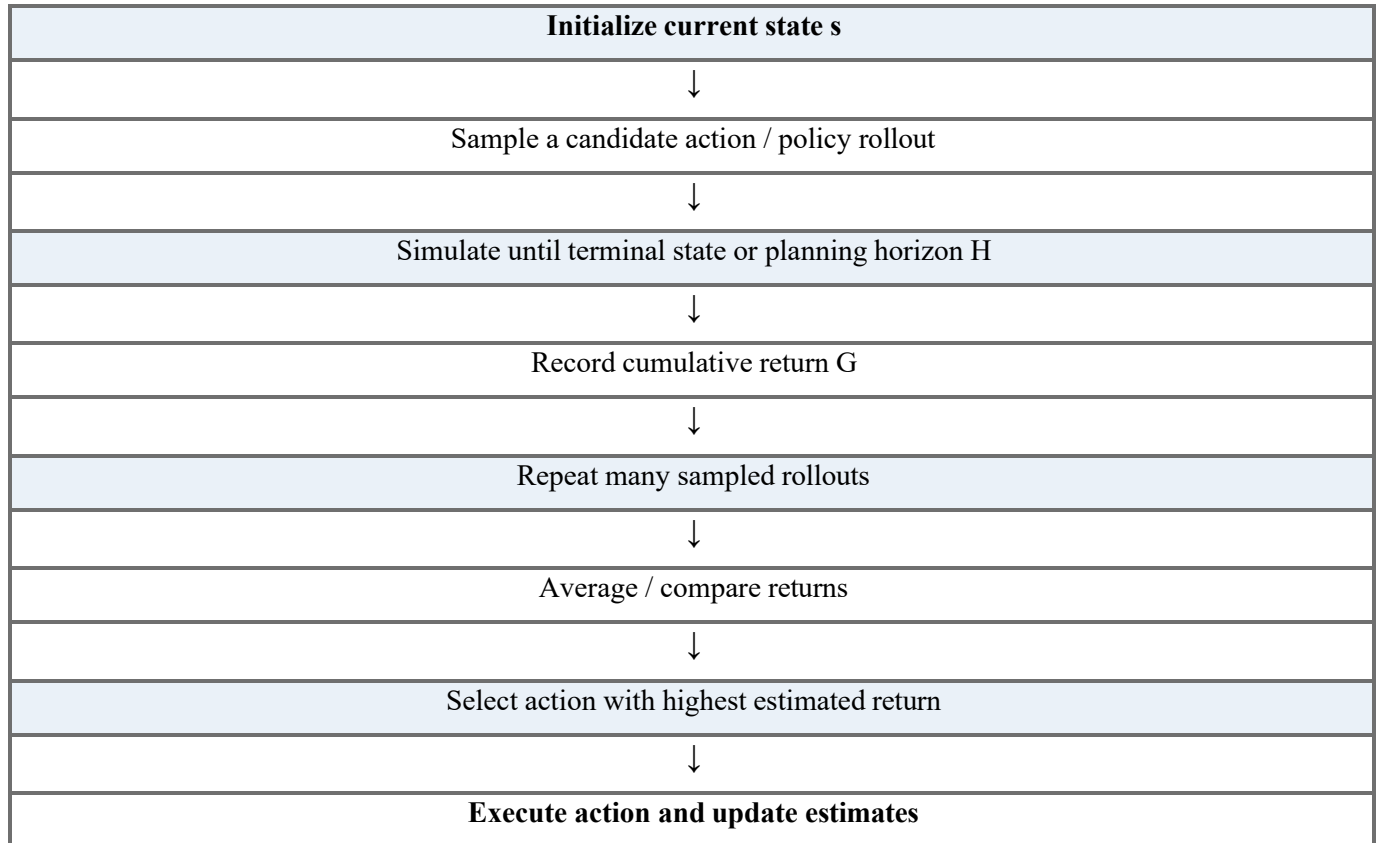
Answer:



REAL ENVIRONMENT → NEW STATE

14. Design a flowchart explaining Monte Carlo sampling for planning in Reinforcement Learning.

Answer:



15. Draw a concept map comparing exhaustive search planning and sampling-based planning techniques.

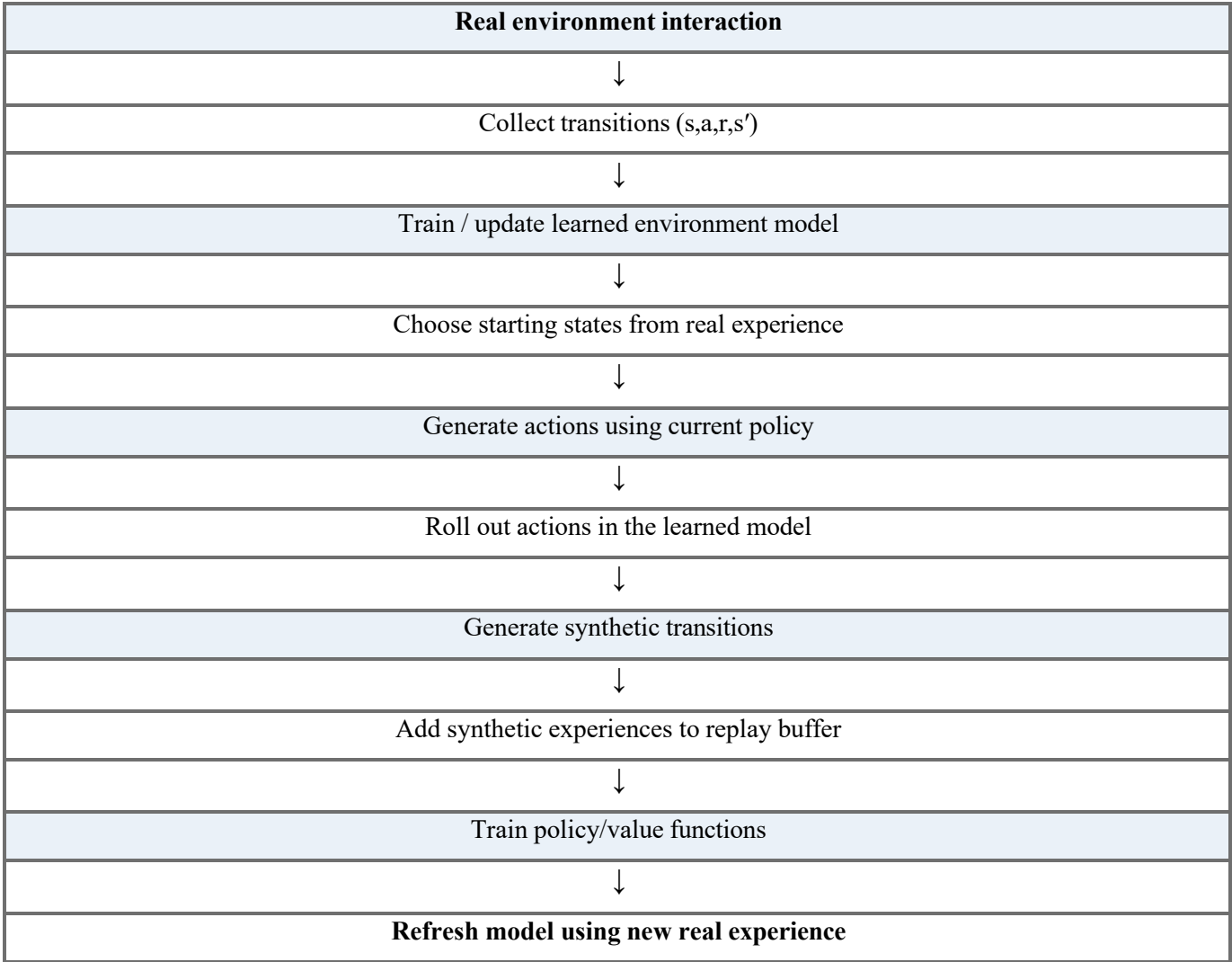
Answer:

EXHAUSTIVE SEARCH	SAMPLING-BASED PLANNING
Explores all feasible branches within a defined search space.	Explores a selected set of sampled trajectories.
Can provide an exact best choice within the searched space.	Provides an estimate dependent on sample quality.
Computational cost grows rapidly with branching factor.	Cost can be controlled by number of samples.
Poor scalability for large action spaces.	Generally more scalable for large spaces.
High coverage but high computation.	Lower coverage but practical computation.

Model-Based Data Generation

16. Draw a flowchart illustrating synthetic data generation using a learned environment model.

Answer:



17. Prepare a concept map showing the relationship among environment model, generated experiences, replay buffer, and policy learning.

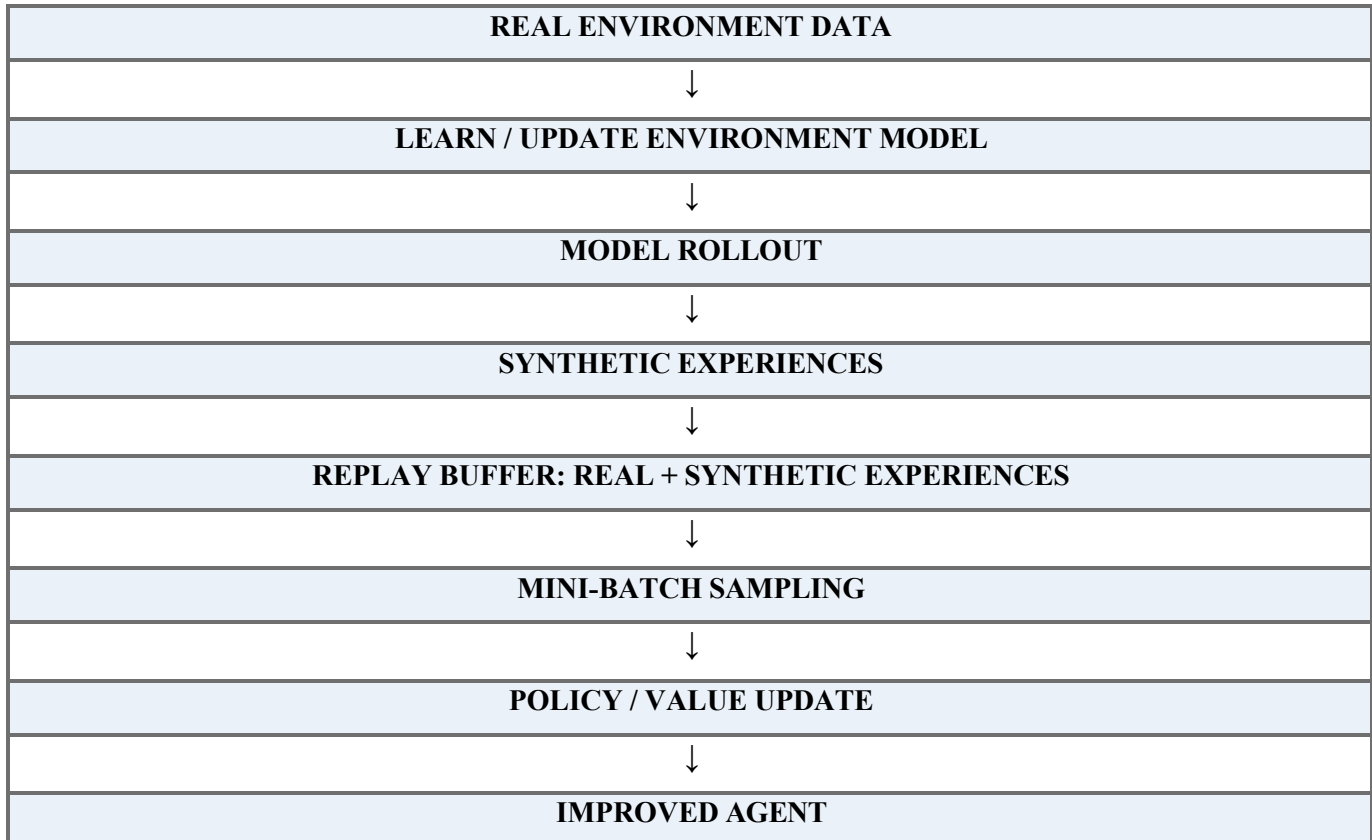
Answer:

MODEL-GENERATED EXPERIENCE REPLAY	
Environment Model	Predicts next states and rewards.
Generated Experiences	Synthetic (s,a,r,s') transitions produced by model rollouts.
Replay Buffer	Stores real and generated transitions for repeated training.
Policy Learning	Samples experiences and updates policy/value

	parameters.
Feedback Loop	Improved policy generates new rollouts while real data corrects model errors.

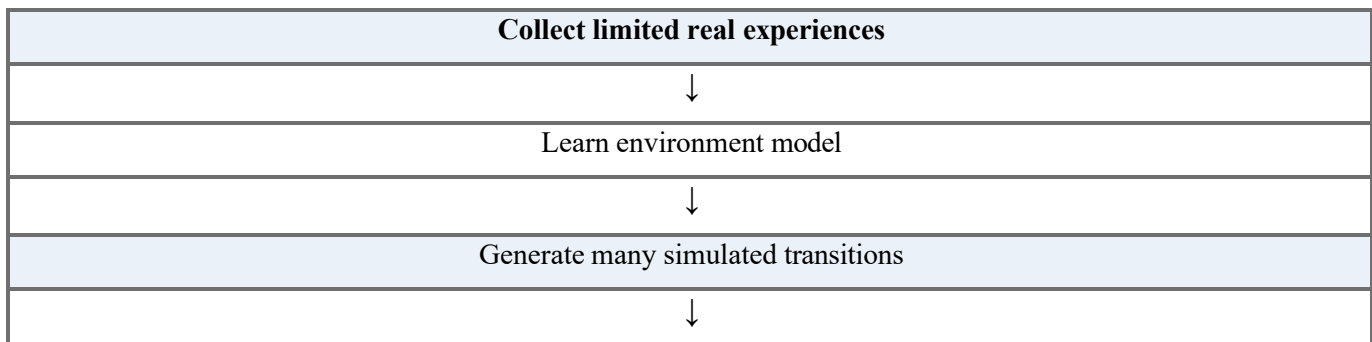
18. Draw a block diagram explaining model-generated experience replay in Reinforcement Learning.

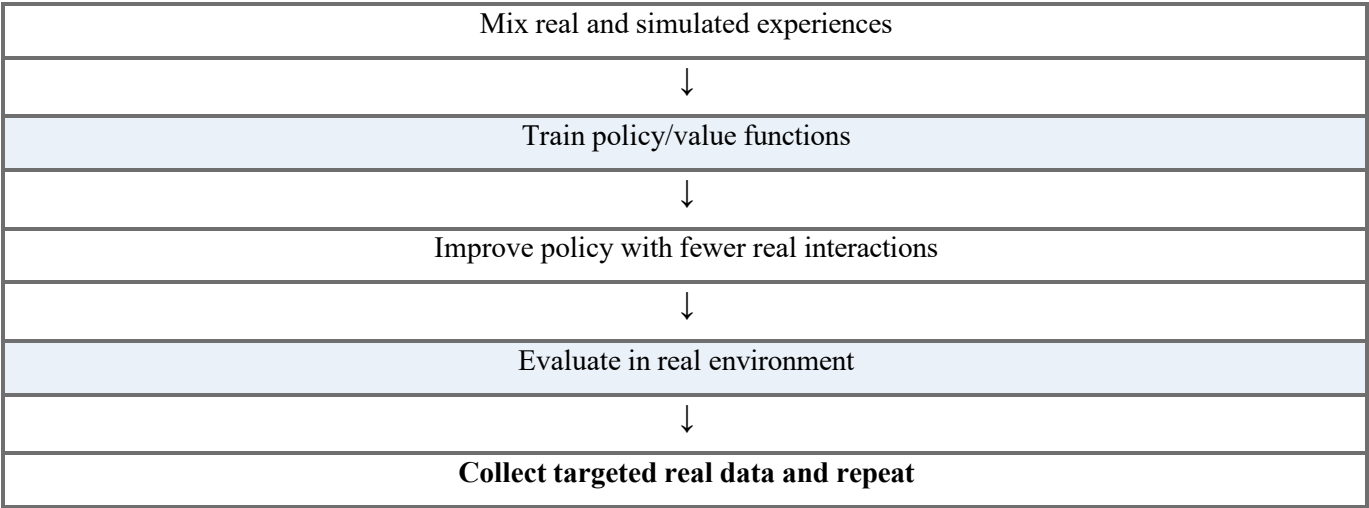
Answer:



19. Design a flowchart showing how simulated experiences improve sample efficiency during training.

Answer:





20. Draw a concept map illustrating the advantages and limitations of model-generated training data.

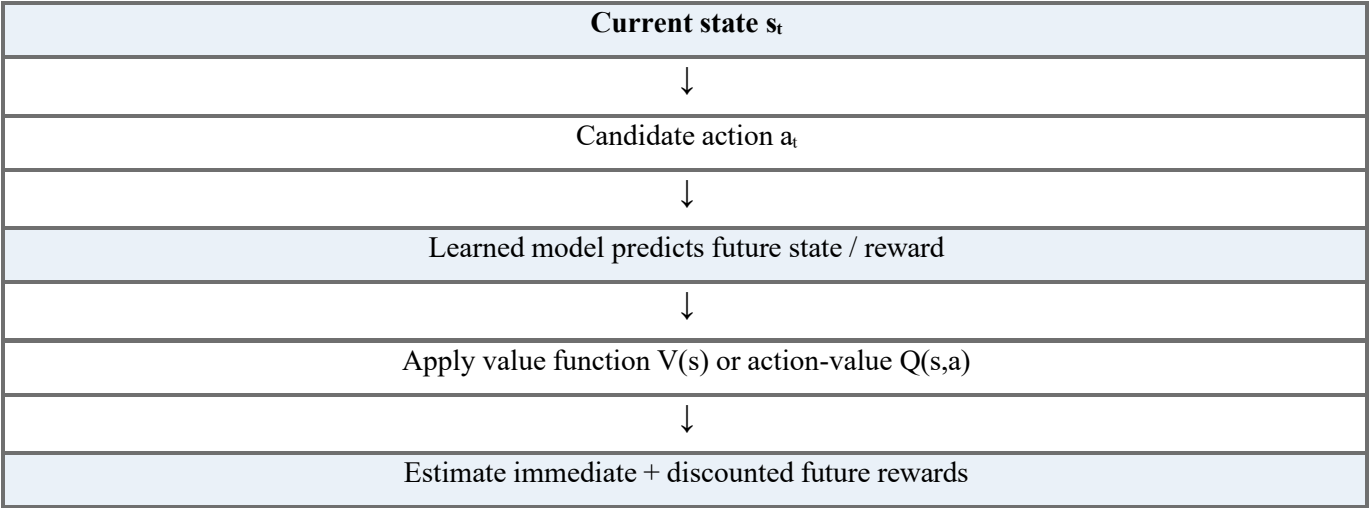
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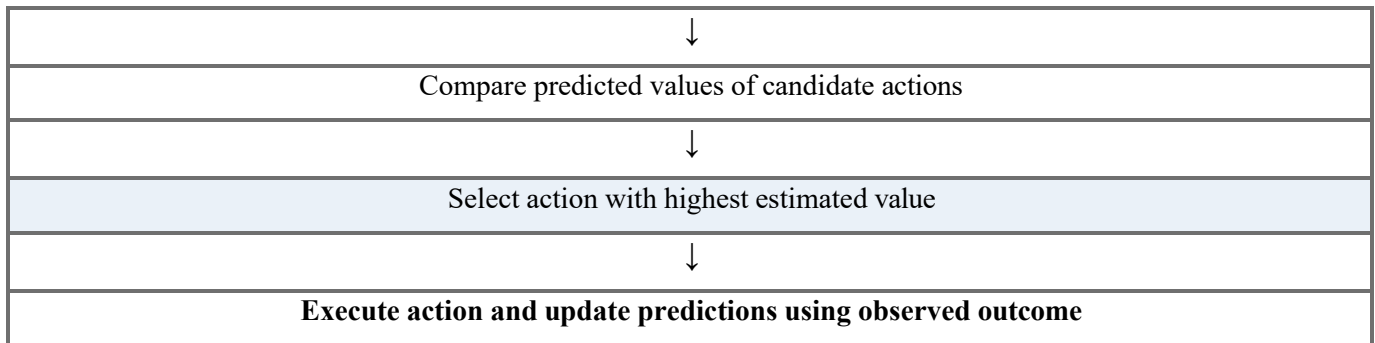
ADVANTAGES	LIMITATIONS
More training data from limited real experience.	Model bias can create inaccurate synthetic data.
Improves sample efficiency.	Prediction errors can compound over long rollouts.
Allows safer exploration in simulation.	Distribution shift may make synthetic data unreliable.
Supports planning without constant real interaction.	Model training and rollout add computational cost.

Value-Equivalence Prediction and Policy Optimization

21. Draw a flowchart explaining value-equivalence prediction for estimating future rewards.

Answer:





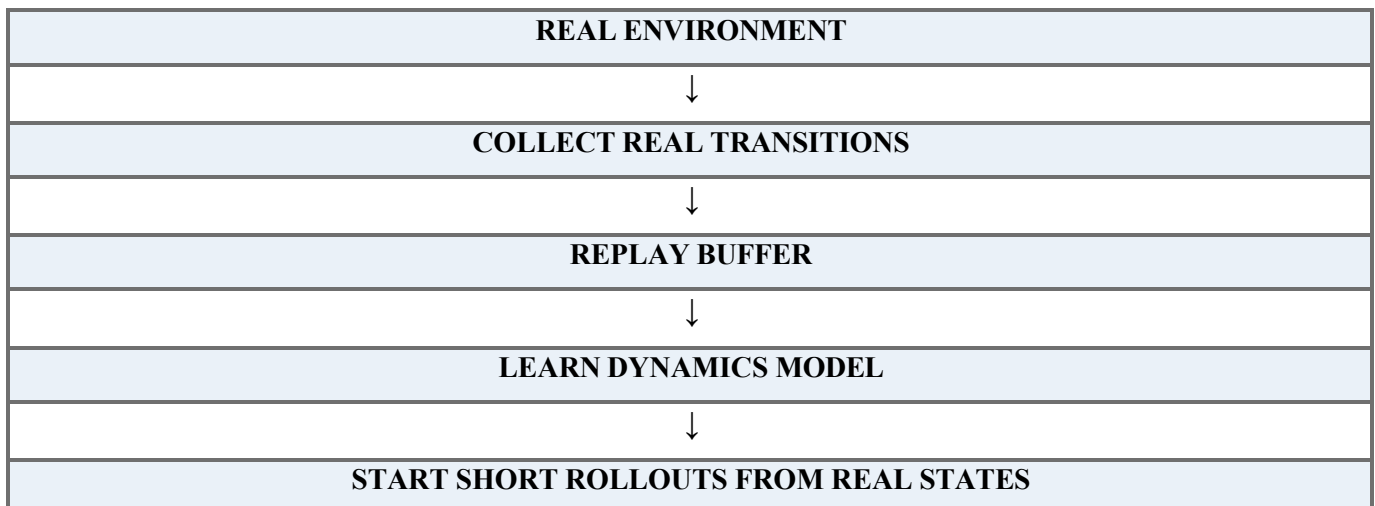
22. Prepare a concept map showing the relationship among value functions, reward prediction, policy evaluation, and policy improvement.

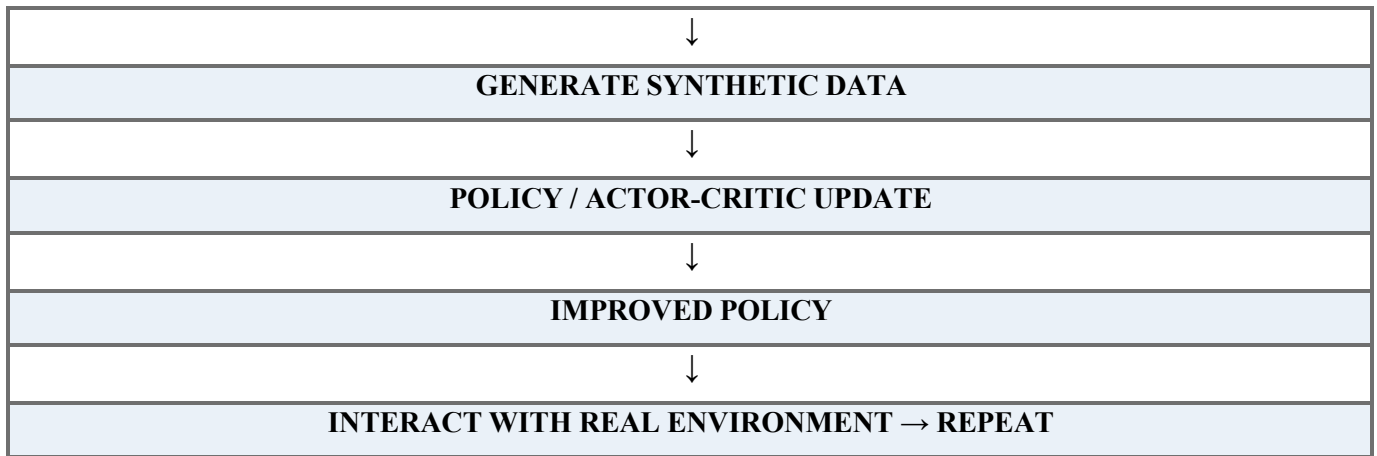
Answer:

VALUE FUNCTIONS AND POLICY OPTIMIZATION	
Reward Prediction	Estimates immediate reward and contributes to future return.
Value Function $V(s)$	Expected return from state s under a policy.
Action-Value $Q(s,a)$	Expected return for taking action a in state s .
Policy Evaluation	Uses value estimates to measure the current policy.
Policy Improvement	Selects actions with better estimated values.
Updated Policy	Generates new experience for another evaluation cycle.

23. Draw a block diagram illustrating the complete Model-Based Policy Optimization (MBPO) framework.

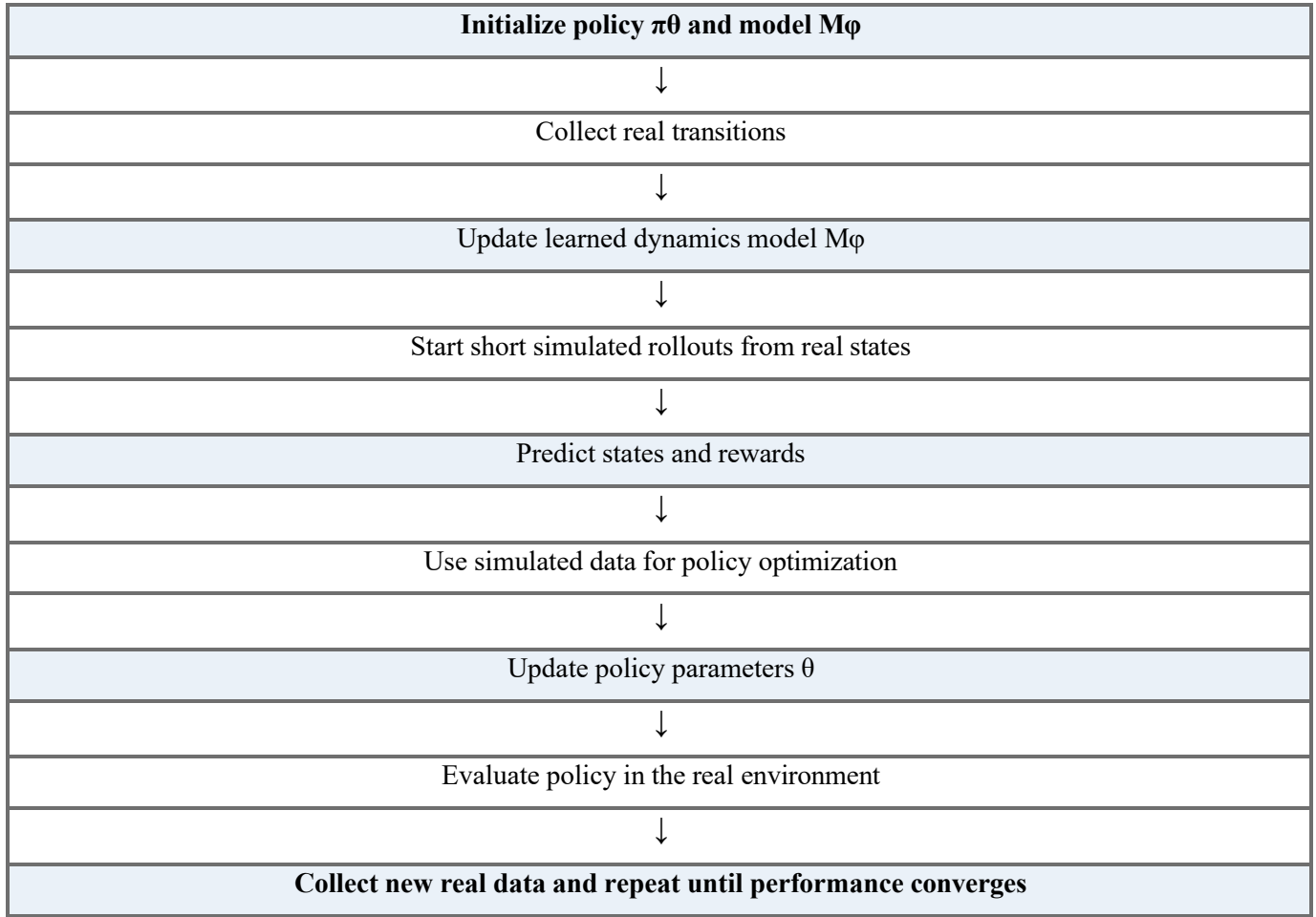
Answer:





24. Design a flowchart explaining iterative policy optimization using learned environment dynamics.

Answer:



25. Draw a comprehensive concept map integrating Model-Based RL, Analytic Gradient Computation, Sampling-Based Planning, Model-Based Data Generation, Value-Equivalence

Prediction, and Model-Based Policy Optimization into a unified Reinforcement Learning framework.

Answer:

UNIFIED MODEL-BASED REINFORCEMENT LEARNING FRAMEWORK	
Model-Based RL	Learns/uses environment dynamics for prediction and decision-making.
Analytic Gradient Computation	Computes policy gradients for parameter optimization.
Sampling-Based Planning	Samples future trajectories and evaluates predicted returns.
Model-Based Data Generation	Creates synthetic experiences through model rollouts.
Value-Equivalence Prediction	Uses value estimates to predict future return and compare actions.
Model-Based Policy Optimization	Combines model learning, synthetic data and policy updates.
Unified Loop	Environment → real data → learned model → planning/data/value prediction → policy optimization → improved policy → environment.

Code of Conduct:

I **Sai Ramana D (192325158L)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **Sai Ramana**

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation